

AI RECRUITER ENGINE – BACKEND ARCHITECTURE MAP

(100% User Story Coverage – Clean & Focused)

Note

This is a high-level backend system map. Final scope, timelines, and cost depend on phase prioritization.

1. CORE SERVICES

A. API Gateway

- Request routing
 - Authentication & Authorization
 - Rate limiting
 - Request logging
 - CORS handling
 - Integrity data encryption middleware
 - Job ad analytics tracking
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B. Authentication Service

- User login / logout
 - JWT token generation
 - Role-based access control (HR, Recruiter, Candidate, Interviewer, Manager)
 - Session management
 - Password reset
 - Consent token generation (Interview Integrity)
 - SSO support (optional)
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2. BUSINESS LOGIC SERVICES

A. Vacancy Service (JD-Driven)

Endpoints

POST /api/vacancies

GET /api/vacancies

GET /api/vacancies/{id}

PUT /api/vacancies/{id}

PATCH /api/vacancies/{id}/status

DELETE /api/vacancies/{id}

GET /api/vacancies/{id}/jd

GET /api/vacancies/{id}/pipeline

GET /api/vacancies/{id}/analytics

Business Logic

- Enforce approved JD linkage (mandatory)
 - Track JD version for audit
 - Vacancy lifecycle rules (Draft/Open/Closed)
 - Prevent deletion if candidates exist
 - Trigger job ad generation
 - Track vacancy KPIs
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B. Job Advertisement Service

Endpoints

POST /api/job-ads/generate

GET /api/job-ads/{vacancyId}

PUT /api/job-ads/{vacancyId}

POST /api/job-ads/{vacancyId}/publish

GET /api/job-ads/{vacancyId}/performance

Business Logic

- Generate ads from JD (AI-powered)
 - SEO optimization
 - Multi-channel publishing (LinkedIn, Indeed, Career Portal)
 - Performance & conversion tracking
 - Template versioning
-

C. Candidate Service

Endpoints

POST /api/candidates

POST /api/candidates/bulk-upload

POST /api/candidates/apply/{vacancyId}
GET /api/candidates
GET /api/candidates/{id}
GET /api/vacancies/{id}/candidates
GET /api/candidates/{id}/ranked
POST /api/candidates/{id}/move-stage

Business Logic

- CV parsing (AI)
 - Competency extraction
 - JD-based scoring & ranking
 - Pipeline stage management
 - Source tracking (job board / referral / internal)
-

D. Interview Service (AI + Traditional)

Endpoints

POST /api/interviews/questions/generate
POST /api/interviews/avatar/invite
POST /api/interviews/traditional/schedule
POST /api/interviews/{id}/start
POST /api/interviews/{id}/responses
POST /api/interviews/{id}/feedback
POST /api/interviews/{id}/calibrate
GET /api/interviews/{id}/analysis
GET /api/interviews/{id}/video

Business Logic

- JD-based question generation
- Difficulty auto-adjustment from JD level
- Interview state machine
- Traditional interviewer feedback collection
- Bias detection & calibration
- Secure video handling

E. Interview Integrity Service

Endpoints

POST /api/integrity/configure/{interviewId}

POST /api/integrity/signals/capture

POST /api/integrity/analyze/{interviewId}

GET /api/integrity/report/{interviewId}

DELETE /api/integrity/data/{interviewId}

Business Logic

- Eye gaze & attention capture
 - Behavioral anomaly detection
 - Context-aware scoring (question difficulty)
 - Integrity risk calculation
 - Consent enforcement & GDPR deletion
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F. Interview Scheduling Service

Endpoints

GET /api/scheduling/interviewers

GET /api/scheduling/slots/check

POST /api/scheduling/slots/book

POST /api/scheduling/reminders/send

Business Logic

- Interviewer availability
 - Calendar sync (Google / Outlook)
 - Slot conflict detection
 - Automatic reminders & no-show tracking
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G. Business Rules Engine

Endpoints

POST /api/rules/validate

GET /api/rules/{category}

POST /api/rules/exceptions/request

Business Logic

- Centralized eligibility rules
 - Promotion / mobility / succession validation
 - Rule priority & conflict detection
 - Exception workflow
 - Audit trail
-

H. Decision Service

Endpoints

GET /api/decisions/recommendations/{vacancyId}

POST /api/decisions/{id}/approve

POST /api/decisions/{id}/override

GET /api/decisions/{id}/history

Business Logic

- Combine CV + Interview + Integrity scores
 - JD-weighted scoring
 - Rules validation before recommendation
 - Human override with justification
 - Time-to-hire & cost tracking
-

I. Internal Mobility Service

Endpoints

POST /api/mobility/scan

GET /api/mobility/{roleId}/matches

POST /api/mobility/{employeeId}/apply

Business Logic

- JD-based role matching
- Skill gap analysis
- Eligibility rules enforcement
- Manager endorsement

J. Succession Planning Service

Endpoints

POST /api/succession/roles

POST /api/succession/{roleId}/identify

GET /api/succession/matrix

Business Logic

- JD as benchmark
 - Readiness calculation (Ready / 6-12 / 12+)
 - Development tracking
 - Coverage validation
-

K. Promotion Service

Endpoints

POST /api/promotions/initiate

GET /api/promotions/eligible

POST /api/promotions/{employeeId}/evaluate

POST /api/promotions/{id}/approve

Business Logic

- Eligibility rules (tenure, performance)
 - JD progression comparison
 - Promotion scoring (AI)
 - Exception handling
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3. INTEGRATION SERVICES

A. JD Integration Service

- Fetch approved JDs
 - Competencies & proficiency levels
 - JD versioning
 - Career ladder mapping
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B. AI Proxy Service

- CV parsing
 - Candidate scoring vs JD
 - Question generation
 - Interview & integrity analysis
 - Promotion & mobility scoring
-

C. Notification Service

- Interview invitations
 - Consent reminders
 - Interview reminders
 - Offer / rejection emails
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D. Document & Media Service

- CV & video storage
 - Secure streaming
 - Transcription
 - Report generation (PDF)
 - GDPR deletion
-

4. DATABASE (SUMMARY)

- Vacancies (JD-linked)
 - Job_Advertisements
 - Candidates
 - Candidate_Vacancy
 - Interviews
 - Interview_Feedback
 - Interview_Integrity_Signals
 - Hiring_Decisions
 - Business_Rules
 - Rule_Exceptions
 - Internal_Mobility
 - Succession_Plans
 - Promotions
 - Audit_Logs
-

DATABASE ARCHITECTURE

1. Core Tables

A. Vacancies

- id (PK)
 - jd_id (FK to JD Engine)
 - jd_version
 - title
 - role_level
 - department
 - employment_type (Full-Time / Part-Time / Contract)
 - location
 - status (Draft / Open / Closed)
 - job_ad_status (Draft / Published / Closed)
 - is_internal (boolean)
 - created_by (FK to Users)
 - created_at
 - updated_at
 - closed_at
-

B. Job_Advertisements

- id (PK)
- vacancy_id (FK)
- generated_content (TEXT)
- template_id (FK, nullable)
- tone (Professional / Friendly / Bold)
- length (Short / Medium / Detailed)
- seo_keywords (JSON)
- company_branding (JSON)

- status (Draft / Published / Closed)
 - generated_at
 - published_at
 - closed_at
-

C. Job_Ad_Publications

- id (PK)
 - job_ad_id (FK)
 - platform (LinkedIn / Indeed / CareerPortal / Custom)
 - platform_job_id
 - posted_at
 - expires_at
 - budget_allocated
 - status (Active / Paused / Closed)
-

D. Job_Ad_Analytics

- id (PK)
 - publication_id (FK)
 - date
 - views
 - clicks
 - applications
 - conversion_rate
 - cost_per_application
 - source_quality_score
-

E. Candidates

- id (PK)
- name
- email

- phone
 - source (LinkedIn / Indeed / Referral / Direct / Internal)
 - source_detail (JSON)
 - cv_file_path
 - parsed_data (JSON)
 - competencies_extracted (JSON)
 - pipeline_stage (Applied / Screened / Interviewed / Offered / Hired / Rejected)
 - created_at
 - updated_at
-

F. Candidate_Vacancy (Junction Table)

- id (PK)
 - candidate_id (FK)
 - vacancy_id (FK)
 - application_source (FK to Job_Ad_Publications)
 - cv_score
 - interview_score
 - traditional_interview_score
 - integrity_score (nullable)
 - combined_score
 - jd_match_percentage
 - ai_reasoning (JSON)
 - jd_competency_gaps (JSON)
 - rank_position
 - status (New / Shortlisted / Interviewed / Rejected / Hired)
 - applied_at
 - last_updated
-

G. Candidate_Communication_Log

- id (PK)

- candidate_id (FK)
 - vacancy_id (FK)
 - communication_type (Email / SMS / Call / In-App)
 - direction (Inbound / Outbound)
 - subject
 - content (TEXT)
 - status (Sent / Delivered / Read / Failed)
 - sent_by (FK to Users)
 - sent_at
-

H. Interviews

- id (PK)
- candidate_id (FK)
- vacancy_id (FK)
- type (AI_Avatar / Traditional)
- interview_stage (Screening / Technical / HR / Manager / Panel)
- status (Scheduled / In_Progress / Paused / Completed / Cancelled / No_Show)
- integrity_monitoring_enabled (boolean)
- consent_given (boolean)
- consent_timestamp
- scheduled_at
- started_at
- completed_at
- meeting_link
- location
- video_file_path
- transcript_file_path
- analysis_data (JSON)
- integrity_analysis_data (JSON)
- created_at

- updated_at
-

I. Interview_Interviewers

- id (PK)
 - interview_id (FK)
 - interviewer_id (FK to Users)
 - role (Primary / Panel / Observer)
 - calendar_event_id
 - attendance_status (Confirmed / Attended / No_Show)
 - feedback_submitted (boolean)
 - assigned_at
-

J. Interview_Questions

- id (PK)
 - interview_id (FK)
 - vacancy_id (FK)
 - question_text
 - competency_mapped
 - jd_competency_id
 - difficulty_level (Junior / Mid / Senior)
 - expected_response_time_seconds
 - question_type (Technical / Behavioral / Situational)
 - order_sequence
 - generated_at
-

K. Interview_Responses

- id (PK)
- interview_id (FK)
- question_id (FK)
- response_text (transcript)

- video_timestamp_start
 - video_timestamp_end
 - score
 - jd_competency_score
 - ai_feedback (JSON)
 - created_at
-

L. Interview_Feedback

- id (PK)
 - interview_id (FK)
 - interviewer_id (FK)
 - overall_rating (1-5)
 - recommendation (Strong_Hire / Hire / Maybe / No_Hire / Strong_No)
 - competency_ratings (JSON)
 - strengths (TEXT)
 - concerns (TEXT)
 - red_flags (TEXT)
 - cultural_fit_rating
 - communication_rating
 - technical_rating
 - open_feedback (TEXT)
 - submitted_at
-

M. Interview_Feedback_Calibration

- id (PK)
- interview_id (FK)
- calibration_session_date
- original_scores (JSON)
- calibrated_scores (JSON)
- consensus_rating

- discrepancies_noted (TEXT)
 - calibration_notes (TEXT)
 - calibrated_by (FK to Users)
 - calibrated_at
-

N. Interview_Integrity_Signals

- id (PK)
 - interview_id (FK)
 - question_id (FK)
 - timestamp
 - eye_gaze_x
 - eye_gaze_y
 - on_screen (boolean)
 - look_away_duration_ms
 - attention_score
 - head_pose_angles (JSON)
 - anomaly_detected (boolean)
 - anomaly_type
 - question_difficulty_context
 - baseline_deviation_percentage
 - created_at
-

O. Interview_Integrity_Reports

- id (PK)
- interview_id (FK)
- overall_integrity_score
- risk_level (Low / Medium / High)
- anomaly_count
- total_look_away_time_seconds
- attention_percentage

- behavioral_patterns (JSON)
 - context_adjustments (JSON)
 - ai_reasoning (JSON)
 - generated_at
-

P. Candidate_Consent_Records

- id (PK)
 - candidate_id (FK)
 - interview_id (FK)
 - consent_type (Integrity_Monitoring / Data_Processing / Communication)
 - consent_given (boolean)
 - consent_text_shown (TEXT)
 - ip_address
 - user_agent
 - timestamp
 - withdrawn_at
-

Q. Hiring_Decisions

- id (PK)
- vacancy_id (FK)
- candidate_id (FK)
- recommendation_type (Hire / Reject / Hold)
- cv_score
- interview_score
- traditional_interview_score
- integrity_score
- combined_score
- jd_match_percentage
- business_rules_passed (boolean)
- failed_rules (JSON)

- ai_reasoning (JSON)
 - hr_decision (Approved / Overridden)
 - override_reason
 - cost_per_hire
 - time_to_hire_days
 - decided_by (FK to Users)
 - decided_at
-

R. Internal_Mobility

- id (PK)
 - employee_id (FK)
 - vacancy_id (FK)
 - target_jd_id (FK)
 - current_jd_id (FK)
 - application_type (Self_Nomination / Manager_Recommendation)
 - role_fit_score
 - jd_competency_match (JSON)
 - skill_gaps (JSON)
 - gap_closure_timeline_months
 - eligibility_status (Eligible / Not_Eligible / Exception_Granted)
 - failed_rules (JSON)
 - manager_endorsement (boolean)
 - endorsed_by (FK to Users)
 - endorsed_at
 - recommendation
 - status (Applied / Under_Review / Approved / Rejected)
 - created_at
 - updated_at
-

S. Succession_Plans

- id (PK)
 - critical_role_id
 - target_jd_id (FK)
 - successor_employee_id (FK)
 - readiness_level (Ready / 6-12mo / 12+mo)
 - jd_competency_gap_percentage
 - competency_data (JSON)
 - development_plan (JSON)
 - business_rules_validated (boolean)
 - validation_notes
 - created_at
 - updated_at
-

T. Promotions

- id (PK)
 - employee_id (FK)
 - promotion_cycle_id (FK)
 - current_jd_id (FK)
 - target_jd_id (FK)
 - current_role
 - proposed_role
 - promotion_score
 - jd_competency_delta (JSON)
 - eligibility_status (Eligible / Not_Eligible / Exception_Granted)
 - failed_rules (JSON)
 - justification (JSON)
 - status (Pending / Approved / Rejected)
 - decided_by (FK to Users)
 - decided_at
-

U. Promotion_Cycles

- id (PK)
 - cycle_name
 - start_date
 - end_date
 - status (Open / Closed)
 - total_eligible_employees
 - total_promoted
 - created_by (FK to Users)
 - created_at
-

V. JD_Competency_Mappings

- id (PK)
 - jd_id (FK to JD Engine)
 - competency_id
 - competency_name
 - proficiency_level (Junior / Mid / Senior / Expert)
 - is_critical (boolean)
 - weight_percentage
 - created_at
-

W. Integrity_Baselines

- id (PK)
- role_level
- question_difficulty
- avg_attention_score
- avg_look_away_duration_ms
- anomaly_threshold_percentage
- sample_size
- last_updated

2. Supporting Tables

X. Users

- id (PK)
- username
- email
- password_hash
- role (HR / Recruiter / Candidate / Interviewer / Manager)
- department
- is_active
- calendar_connected (boolean)
- calendar_provider
- availability_preferences (JSON)
- created_at

Y. Interviewer_Profiles

- id (PK)
- user_id (FK)
- interview_types (JSON)
- competency_expertise (JSON)
- max_interviews_per_week
- avg_rating_given
- total_interviews_conducted
- feedback_quality_score
- last_calibration_date

Z. Interviewer_Availability

- id (PK)
- interviewer_id (FK)

- day_of_week
 - start_time
 - end_time
 - is_available
 - buffer_minutes_between
-

AA. Calendar_Events

- id (PK)
 - interview_id (FK)
 - external_calendar_id
 - calendar_provider
 - event_link
 - created_at
 - last_synced_at
-

AB. Audit_Logs

- id (PK)
 - user_id (FK)
 - action
 - entity_type
 - entity_id
 - jd_version_referenced
 - changes (JSON)
 - ip_address
 - user_agent
 - timestamp
-

AC. Notifications

- id (PK)
- recipient_id (FK)

- type (Email / SMS / In_App)
 - category (Interview / Application / Decision / Reminder)
 - subject
 - content
 - status (Sent / Failed / Pending / Delivered / Read)
 - delivery_timestamp
 - read_timestamp
 - sent_at
-

AD. Integrity_Data_Access_Log

- id (PK)
 - user_id (FK)
 - interview_id (FK)
 - access_type (View / Export / Delete)
 - purpose
 - timestamp
-

AE. System_Configuration

- id (PK)
 - config_key
 - config_value (JSON)
 - config_category
 - description
 - is_active
 - updated_by (FK to Users)
 - updated_at
-
-

USER STORY COVERAGE

- ✓ EPIC 1 – Vacancy Management
- ✓ EPIC 2 – Candidate Ingestion & Scoring
- ✓ EPIC 3 – Candidate Transparency
- ✓ EPIC 4 – Interview Management
- ✓ EPIC 5 – Interview Integrity
- ✓ EPIC 6 – Hiring Decisions
- ✓ EPIC 7 – Internal Mobility
- ✓ EPIC 8 – Succession Planning
- ✓ EPIC 9 – Promotion Management